

Math 170S - Homework 5 - due Feb 20

Problem 1. 7.2-2

Problem 2. 7.3-10

Problem 3. 7.3-12

Problem 4. 7.4-7

Problem 5. Let $\bar{X}$ be the mean of a random sample of size n from $N(\mu_X, 4^2)$, $\bar{Y}$ be the mean of a random sample of size n from $N(\mu_Y, 3^2)$, all X_i 's and Y_i 's are independent. What n should be enough to ensure that

$$\mathbb{P}(\mu_X - \mu_Y \leq \bar{X} - \bar{Y} + 0.5) \geq 0.9?$$

Problem 6. 8.1-6

Problem 7. 8.1-12